

Deciphering the Platinum Electrolyte Interface in Aqueous Solution (Report)

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In this study, we calculated the lattice parameters and bulk modulus for platinum (Pt) bulk. The bulk structure was constructed using the Python library *pymatgen*[1] for a periodic Pt slab ($2 \times 2 \times 3$) with 48 atoms. The cell volume was varied through a lattice constant ranging from 3.90 to 4.10 Å. The DFT calculations were performed using CP2K/QUICKSTEP. The 5d and 6s electrons of Pt were treated as valence electrons, with the remaining core electrons represented by Goedecker-Teter-Hutter pseudopotentials. [2] The Gaussian basis sets were double ζ (DZVP-MOLOPT-SR-GTH), and the energy cutoff was varied from 100 Ry to 1000 Ry. We used different functionals to describe exchange-correlation effects and considered the dispersion correction with the Grimme D3 method to the entire bulk (RPBE+VDW), and only to one surface layer (RPBE+VDW/1L).

1 Lattice parameter

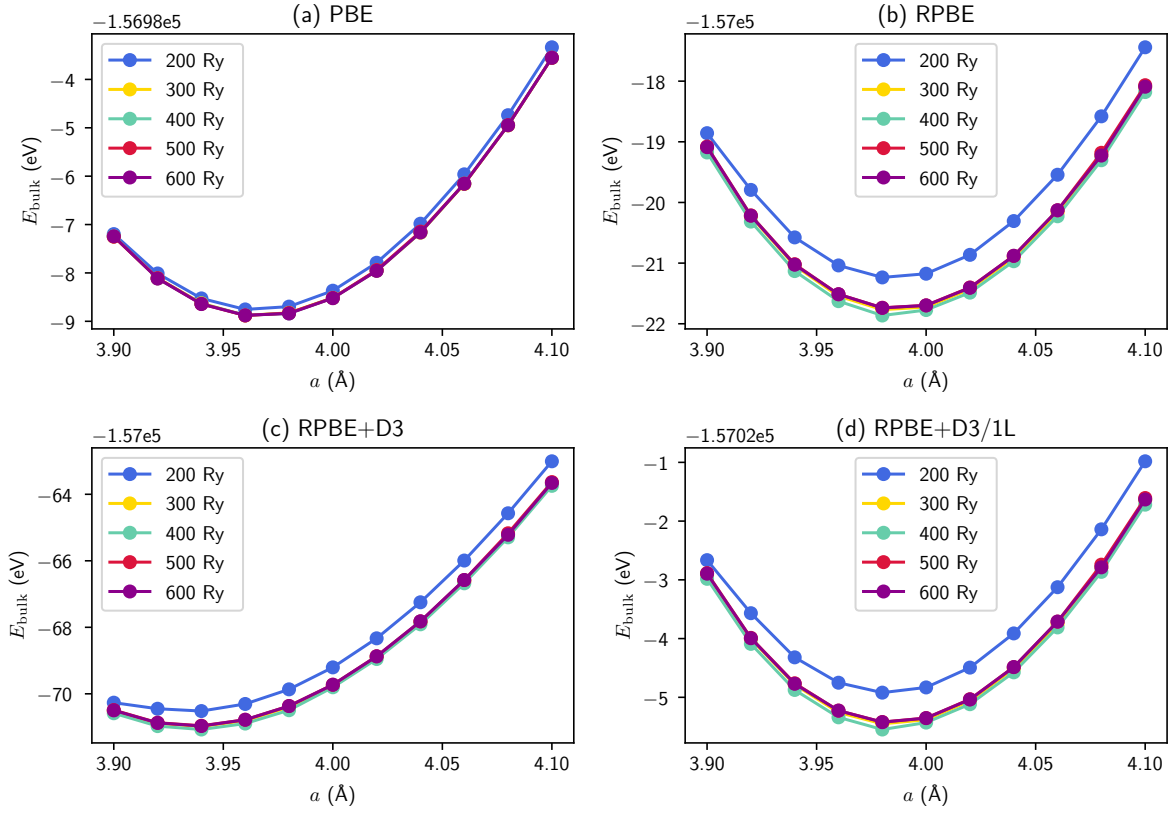


Figure 1: Total energies as a function of lattice parameter a (Å) for Pt (fcc) using different functionals A) PBE, B) RPBE, C) RPBE+VDW (D3 for all Pt), and D) RPBE+VDW/1L (D3 for one Pt layer) with various cutoff energy values.

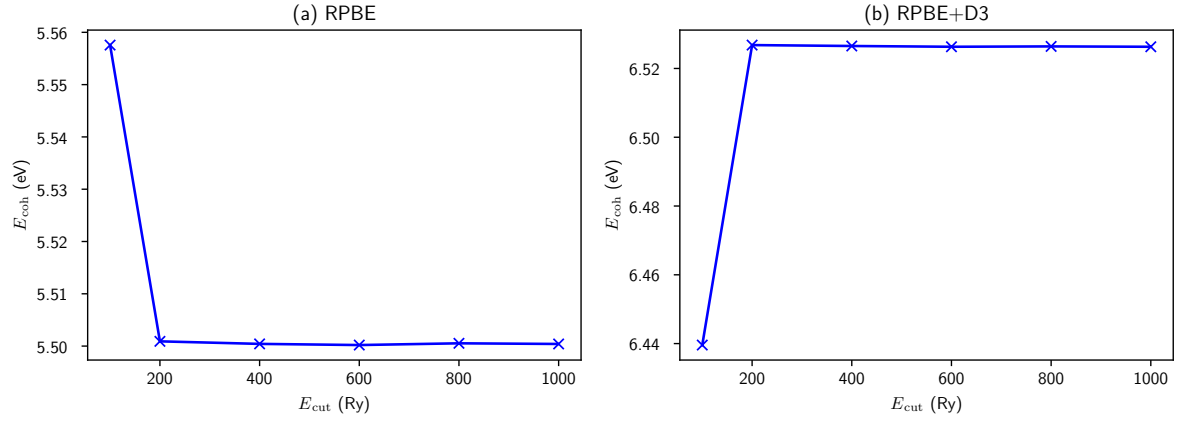


Figure 2: Cohesive energies E_{coh} (eV) as a function of cutoff energy E_{cut} (Ry) for Pt (fcc) using different functionals A) RPBE, B) RPBE+VDW (D3 for all Pt), and C) RPBE+VDW/1L (D3 for one Pt layer).

2 Bulk modulus

The total energies in Fig. 2 were fitted with the Birch-Murnaghan equation of state to obtain the bulk modulus. The Birch-Murnaghan equation of state is given by:

$$P(V) = \frac{3B_0}{2} \left[\left(\frac{V_0}{V} \right)^{7/3} - \left(\frac{V_0}{V} \right)^{5/3} \right] \left\{ 1 + \frac{3}{4}(B'_0 - 4) \left[\left(\frac{V_0}{V} \right)^{2/3} - 1 \right] \right\} \quad (1)$$

Where P is the pressure, V_0 is the reference volume, V is the deformed volume, B_0 is the bulk modulus, and B'_0 is the derivative of the bulk modulus with respect to pressure. The energy associated with this equation of state, $E(V)$, can be obtained by integrating the pressure with respect to the volume:

$$E(V) = E_0 + \frac{9V_0B_0}{16} \left\{ \left[\left(\frac{V_0}{V} \right)^{2/3} - 1 \right]^3 B'_0 + \left[\left(\frac{V_0}{V} \right)^{2/3} - 1 \right]^2 \left[6 - 4 \left(\frac{V_0}{V} \right)^{2/3} \right] \right\} \quad (2)$$

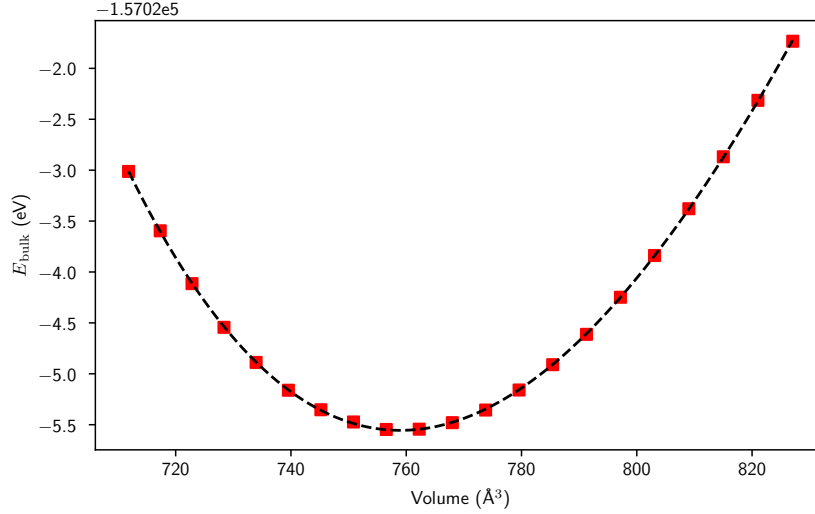


Figure 3: Total energy as a function of volume using RPBE+VDW (1L). The solid blue curve represents the data fitted according to the Birch-Murnaghan equation of state (EoS). The dots represent the actual calculated results with a cutoff value of 500 Ry.

3 Basis set

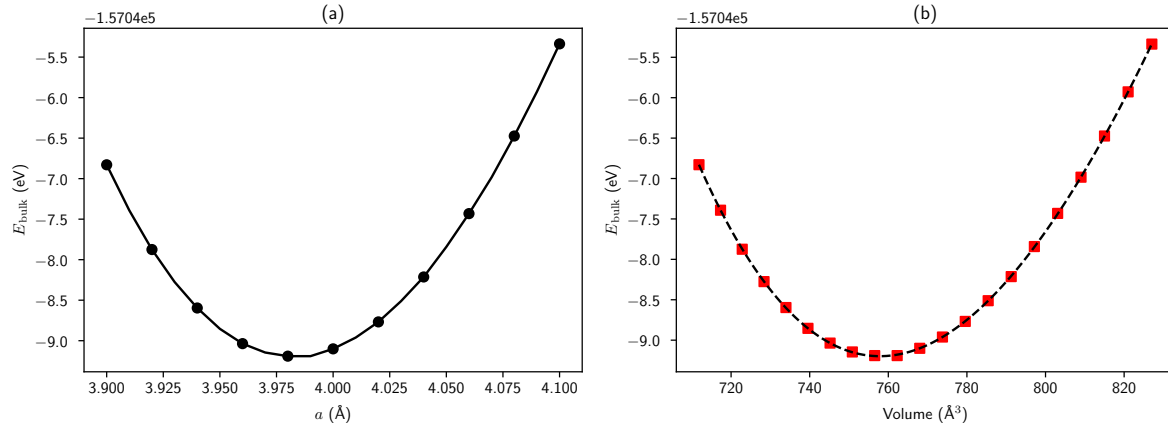


Figure 4: Total energies as a function of A) lattice parameter a (Å) and B) Volume (Å³) for Pt (fcc) using RPBE+VDW/1L functional (D3 for one Pt layer) and GTH-UZH potentials.[2]

4 Results

The results are tabulated in Table 1 using different functionals and experimental values along with the values from previous DFT calculations using VASP by Surendralal[3].

Functional	a (Å) this work	E_{coh} (eV) this work	B_0 (GPa) this work	a (Å) Surendralal[3]	E_{coh} (eV) Surendralal[3]
PBE	3.97		259	3.97	5.56
RPBE	3.99	5.50	241	3.99	4.90
RPBE+VDW	3.94	6.63	251	3.94	5.93
RPBE+VDW (1L)	3.98	5.58	242		
RPBE+VDW (1L) - UZH	3.99		237		
Experimental value[4]	3.92	5.86	278		

Table 1: Calculated lattice parameters and bulk modulus of Pt (fcc) with different functionals at 500 Ry cutoff.

The calculated lattice parameters are higher when compared to the experimental value but agree with the previous DFT calculations (Surendralal.[3]).

References

- [1] Richard Tran, Zihan Xu, Balachandran Radhakrishnan, Donald Winston, Wenhao Sun, Kristin A. Persson, and Shyue Ping Ong. Surface energies of elemental crystals. *Scientific Data*, 3(1):160080, September 2016.
- [2] S. Goedecker, M. Teter, and J. Hutter. Separable dual-space gaussian pseudopotentials. *Phys. Rev. B*, 54:1703–1710, Jul 1996.
- [3] Sudarsan Surendralal. *Development of an ab initio computational potentiostat and its application to the study of Mg corrosion*. PhD thesis, Ruhr-Universität Bochum, 2019.
- [4] C. Kittel. *Introduction to Solid State Physics*. Wiley, 2004.